PACKING $k^2 - \lceil\sqrt{32k}\,\rceil - 1$ UNIT SQUARES IN A SQUARE OF SIDE LESS THAN k

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ABSTRACT. Let $s(N)$ be the side of the smallest square containing N unit squares with disjoint interiors, and let $c^*(k)$ be the largest c with $s(k^2 - c) = k$. We prove $c^*(k) < 4\sqrt{2k} + 1$ for every integer $k \geq 6$, that is, $s(k^2 - \lceil\sqrt{32k}\,\rceil - 1) < k$. The construction is Ryu's L-shaped packing: an aligned grid and two bands of b tilted rows. Both ends of each band are filled with axis-parallel rows, and b is the odd integer nearest $\sqrt{2k}$. The odd width keeps the rounding loss of each filled end below $3/2$. This lowers the constant of the $O(\sqrt{k})$ bound valid for every $k \geq 6$ from 8 (Ryu) to $4\sqrt{2}$. The bound is the smallest explicit one we know of for $42 \leq k < 1.6 \cdot 10^7$, except at five values of k where Ryu has computer-assisted bounds. Exact rational certificates for $k \leq 600$ are checked by two independently written programs.

1. INTRODUCTION

For $x > 0$ let $N(x)$ be the largest number of unit squares that fit, with pairwise disjoint interiors and rotations allowed, in a square of side x. For an integer k, $N(k) = k^2$. Let

$$M(k) = \max\{N : s(N) < k\}, \qquad w(k) = k^2 - M(k), \qquad c^*(k) = \max\{c : s(k^2 - c) = k\}.$$

Then $s(k^2 - c) < k$ exactly when $M(k) \geq k^2 - c$, so $c^*(k) = w(k) - 1$.

Erdős and Graham [EG75] showed that the waste $x^2 - N(x)$ is $O(x^{7/11})$, which makes $w(k)$ sublinear. The best exponent is now $3/5$ [Bui25, McC26] (the earlier claim [CG20] contained an error [AB26]). For integer sides approached from below, Ryu [Ryu26] proved, among other things:

- $c^*(k) \leq 8\lceil\sqrt{k-4}\,\rceil - 1$ for every $k \geq 6$ (his Theorem 1.4), from an explicit L-shaped packing (his Theorem 1.3);
- $c^*(k) < 20.668\, k^{2/5} + 0.623$ for $k \geq 1.6 \cdot 10^7$ (Theorem 1.2), and bounds of order $k^{3/8}$ from $k \geq 2.48 \cdot 10^{12}$ (Theorem 1.1);
- computer-assisted bounds at five values of k between 10^5 and 10^8 (Theorem 1.6).

Lower bounds of order $\log k$, $k^{1/4}$ and $k^{1/3}$ for large k are also due to Ryu [Ryu26b]. Arslanov, Mustafin and Shangitbayev [AMS21] proved $c^*(k) \leq k - 1$ for $k \geq 12$. In an earlier note [Fan26a] we proved $c^*(k) \leq k - 1 - F(k)$ for $k \geq 15$, with $F(k) = 1 + \max\{p + 2q : 16p + 21q \leq k - 15\}$.

Theorem 1. *For every integer $k \geq 6$, $w(k) < 4\sqrt{2k} + 2$. Equivalently $c^*(k) < 4\sqrt{2k} + 1$, so $c^*(k) \leq \lceil\sqrt{32k}\,\rceil$, and $s(n) < k$ for every integer $n \leq k^2 - \lceil\sqrt{32k}\,\rceil - 1$.*

The packing is Ryu's L (his Sections 3–4), and we use his end-region framework (his Section 6): lifted band ends, the far end as the 180° image of the near one, and end fills assembled with the rows. What is new is the fill (axis-parallel rows at every integer level of each end region), the choice of an odd band width, and the estimate of Lemma 2. To keep the note self-contained, we prove everything we use.

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Draft v0.1, 2026-10-24. Code and certificates: [Fan26b], folder `theorem-a/`.

2. The band

Fix an integer $b \geq 3$ and a rational $t = \tan(\theta/2) > 1/b$. Write $c = \cos\theta = (1-t^2)/(1+t^2)$, $s = \sin\theta = 2t/(1+t^2)$, $T = s/c$, $\tau = c/s$, $\sec = 1/c$, and

$$W = bc + s \quad \text{(band width)}, \qquad h = bs + c \quad \text{(row height)}.$$

At $t_0 = 1/b$: $c_0 = (b^2-1)/(b^2+1)$, $s_0 = 2b/(b^2+1)$, $\tau_0 = (b^2-1)/(2b)$, $W_0 = b$, $h_0 = (3b^2-1)/(b^2+1) < 3$. Since $dW/d\theta = c - bs < 0$ for $\tan\theta > 1/b$, and $\theta_0 = 2\arctan(1/b) > \arctan(1/b)$, we have $W < b$ for every $t > 1/b$.

Use band coordinates $X \in [0, W]$ (across) and $Y \in [0, L]$ (along). With (u, v) the coordinates along (c, s) and $(-s, c)$, the rectangle $P = \{0 \leq u \leq b,\ 0 \leq v \leq 1\}$ is the union of the b unit squares $\{i \leq u \leq i+1\}$. *Row* j is P translated so that its corner $u = v = 0$ is at $(s,\ y_0 + j\sec)$, for $0 \leq j < R$.

(R1) A point of P has $X - s = uc - vs \in [-s, bc]$, so every row lies in $0 \leq X \leq W$. *(R2)* The shift $(0, \sec)$ changes v by exactly 1, so in row 0's frame row j lies in $\{j \leq v \leq j+1\}$ and the rows have disjoint interiors. *(R3)* The lowest point of row j is its corner, at height $y_0 + j\sec$, and its highest is at height $y_0 + j\sec + h$. So the rows fit in $[0, L]$ when $y_0 + (R-1)\sec + h \leq L$. Let $R = R(L) = \lfloor (L-h)/\sec \rfloor + 1$, the most rows with $y_0 = 0$, and let $F = L - (R-1)\sec - h \in [0, \sec)$ be the free length.

End regions. With row 0 at lift y, let $\ell(X) = y + (X-s)T$ for $X \geq s$ and $\ell(X) = y + (s-X)/T$ for $X \leq s$. *(E1)* Every row lies in $\{Y \geq \ell(X)\}$: in row 0's frame a point of any row has $u, v \geq 0$, and $Y - y - (X-s)T = vc + vs^2/c \geq 0$ for $X \geq s$, while $Y - y - (s-X)/T = us + uc^2/s \geq 0$ for $X \leq s$. So a unit square in $\{0 \leq X \leq W,\ 0 \leq Y \leq \ell(X)\}$ meets no row in an interior point.

Fill. Let $a_i(y) = \max(0,\ s + (i+1-y)\tau)$ and $m_i(y) = \max(0, \lfloor W - a_i(y) \rfloor)$. Level i holds the m_i unit squares $[W - m_i + q,\ W - m_i + q + 1] \times [i, i+1]$, $0 \leq q < m_i$, and $\mathrm{fill}(y) = \sum_i m_i(y)$. *(E2)* These squares lie under ℓ. If $a_i \geq s$, ℓ increases on $[a_i, W]$ and $\ell(a_i) \geq i+1$. If $a_i < s$, then $i + 1 < y \leq \min \ell$.

(E3) The 180° rotation about $(W/2, L/2)$ maps the rows onto rows with lifts $y_1 + (R-1-j)\sec$, where $y_1 = L - y_0 - (R-1)\sec - h$. So the region over the last row is a rotated end region with lift y_1, and takes the rotated fill. The two fills are disjoint, since each vertical line $X = \text{const}$ meets row 0 and row $R-1$. A band of length L therefore holds

$$C(L) = bR + \mathrm{fill}(y_0) + \mathrm{fill}(y_1)$$

unit squares. Below, $y_0 = 0$ and $y_1 = F$.

3. The packing

Let $S = k - b + W < k$, and suppose $k - b > h$. Put three pieces in $[0, S]^2$:

- the aligned grid $[0, k-b]^2$;
- the band of length S in $[k-b, S] \times [0, S]$ (with $x = k - b + X$, $y = Y$);
- the band of length $k-b$, reflected in $y = x$, in $[0, k-b] \times [k-b, S]$.

They have disjoint interiors. Since $k^2 - (k-b)^2 = bk + b(k-b)$,

$$w(k) \leq b(k-S) + \big[bS - C(S)\big] + \big[b(k-b) - C(k-b)\big]. \tag{1}$$

4. The end lemma

From now on b is odd. Put $\eta = b/2 - \tau$; at t_0, $\eta_0 = 1/(2b)$. Say $t > 1/b$ is *close* if

(a) $\lfloor W - s - \tau \rfloor = (b-1)/2$;
(b) $0 \leq \eta < 1/2$;
(c) $2\tau - bc \geq 0$;
(d) $h < 3$.

At t_0 we have $W_0 - s_0 - \tau_0 = (b-1)/2 + \delta_b$ with $\delta_b = \frac{1}{2} + \frac{1}{2b} - \frac{2b}{b^2+1} \in (0,1)$; here $\delta_b > 0$ because $b^3 - 3b^2 + b + 1 > 0$ for $b \geq 3$. Also $2\tau_0 - bc_0 = (b^2-1)/(b(b^2+1)) > 0$. So (a)–(d) hold strictly at t_0, and by continuity on an interval $(1/b, 1/b + \varepsilon_b)$.

Lemma 2. *If t is close, then* $\mathrm{fill}(0) \geq (b-1)/2$, *and for* $0 \leq y < \sec$,

$$\mathrm{fill}(y) > \tfrac{b-1}{2} + 2z - 1 - 2\eta, \qquad z = W - s - 2\tau + y\tau.$$

Proof. $\mathrm{fill}(0) \geq m_0(0) = \lfloor W - s - \tau \rfloor = (b-1)/2$ by (a). For $0 \leq y < \sec$ and $i \geq 0$, the clipping in a_i is inactive:

$$s + (i + 1 - y)\tau > s + (1 - \sec)\tau = c(1-c)/s \geq 0.$$

So level 0 holds at least $\lfloor z + \tau \rfloor$ squares and level 1 at least $\lfloor z \rfloor$. Since $\tau = \frac{b-1}{2} + \frac{1}{2} - \eta$ with $\frac{b-1}{2} \in \mathbb{Z}$, together they hold at least $\frac{b-1}{2} + \lfloor z + \frac{1}{2} - \eta \rfloor + \lfloor z \rfloor$. For $0 \leq \eta < \frac{1}{2}$ and real z with fractional part f, the sum $\lfloor z \rfloor + \lfloor z + \frac{1}{2} - \eta \rfloor$ is $2z - 2f > 2z - 1 - 2\eta$ if $f < \frac{1}{2} + \eta$, and $2z - 2f + 1 > 2z - 1$ otherwise. □

The odd width matters: it puts the arguments of the two floors half a unit apart. For even b the arguments differ by an integer minus $1/(2b)$, and this pairing gives nothing.

Lemma 3. *If t is close and $L > h$, then with $R = R(L)$, $y_0 = 0$ and $y_1 = F$,*

$$bL - C(L) < bL(1-c) + bch - 2b + 2 - 2(W - s - 2\tau) + 2\eta. \tag{2}$$

At t_0 the right side equals $2bL/(b^2+1) + b + 2 - \kappa_b$, where $\kappa_b = (7b^4+1)/\big(b(b^2+1)^2\big)$.

Proof. $bR = b + bc(L - h - F)$. By Lemma 2, $C(L) > bR + (b-1) + 2(W - s - 2\tau) + 2F\tau - 1 - 2\eta$. The terms in F sum to $F(2\tau - bc) \geq 0$ by (c), so drop them and rearrange. The value at t_0 uses $1 - c_0 = 2/(b^2+1)$, $bc_0h_0 = 3b - b(10b^2+2)/(b^2+1)^2$, $W_0 - s_0 - 2\tau_0 = -(b^2-1)/(b(b^2+1))$ and $2\eta_0 = 1/b$. □

5. Proof of Theorem 1

Let $k \geq 6$ and let b be an odd integer with $|b - \sqrt{2k}| \leq 1$. Then $b \geq 3$. For $k \geq 8$, $k - b \geq k - \sqrt{2k} - 1 \geq 3 > h$ by (d). For $k = 6, 7$ the only choice is $b = 3$, and $k - b \geq 3 > h$. For close t, (1) and (2) with $L = S$ and $L = k - b$ give $k^2 - N(t) < H(t)$, where $N(t)$ is the number of squares packed and $H(t)$ is $b(k - S)$ plus the two right sides of (2).

This holds for each t separately, even though R, F and the floors jump with t. H is continuous, and as $t \to 1/b^+$, $S \to k$ and

$$H(t) \to B(k) = \frac{4kb}{b^2+1} + 2b + 2 + \frac{2}{b^2+1} - 2\kappa_b.$$

Now $4kb/(b^2+1) < 4k/b$, and $4k/b + 2b = 4\sqrt{2k} + \frac{2}{b}(b - \sqrt{2k})^2 \leq 4\sqrt{2k} + \frac{2}{b}$. So

$$B(k) < 4\sqrt{2k} + 2 - 2\Big(\kappa_b - \frac{1}{b} - \frac{1}{b^2+1}\Big),$$

and $b(b^2+1)^2(\kappa_b - \frac{1}{b} - \frac{1}{b^2+1}) = 6b^4 - b^3 - 2b^2 - b > 0$. Hence $B(k) < 4\sqrt{2k} + 2$. Some close t has $H(t) < 4\sqrt{2k} + 2$, and it gives a packing with $S < k$ and $k^2 - N(t) < 4\sqrt{2k} + 2$. □

Remark (Sharpness). For odd $b \geq 5$ and $k = (b^2 - b + 8)/2$, the construction at close t has waste exactly $4b$, which is $4\sqrt{2k} + 2 - 31/(2b) + O(b^{-2})$; for example $k = 59$, $b = 11$, $w = 44$. So the additive constant 2 cannot be lowered for this construction.

6. Checks and certificates

The construction is implemented in exact rational arithmetic (`lband.py` in [Fan26b]).

- *Admissibility.* For each k the program chooses the squeeze $k - S$ as $1/(10^6 k^3)$, divided by 10 until (a)–(d) and $H(t) < 4\sqrt{2k} + 2$ hold. It decides those conditions exactly, so every layout it returns satisfies the theorem's bound by Lemma 3.
- *A fixed squeeze is not enough.* With a fixed squeeze of 10^{-7} the same recipe has waste $1178 = 4\sqrt{2k} + 2$ at $k = 43218$, which violates the strict bound.
- *Every step checked.* Every inequality in the proof was checked exactly for $6 \le k \le 20000$, and the bound for $6 \le k \le 60000$. The least slack $B(k) - w$ is $8019401/78023840599$, at $k = 19705$.
- *Certificates.* Strict certificates (every wall and pair margin positive) of the construction for $6 \le k \le 40$ and $k = 101, 300, 600$ pass our checker and Evan Daniel's [Dan26]. The largest has 359860 squares. A program that is not all-pairs runs both checkers unmodified on overlapping 4×4 windows: two unit squares whose centres are more than $\sqrt{2}$ apart cannot overlap, and both checkers accept such pairs without a separating-axis test.

The construction is not optimized. Searching over b, the number of rows and the split of the free length between the two ends gives waste at most $4\sqrt{2k} + 1.18$ for $15 \le k \le 300$; for example 9946 squares below side 100. A bounded search for better end and corner pieces by local optimization found gains of one or two squares only (at $k = 25, 30, 40$), not growing with b. That is consistent with the heuristic that a boundary between the aligned and the tilted lattice wastes about $1/2$ per unit length.

7. Comparison

The integer form of the theorem is $c^*(k) \le \lceil \sqrt{32k} \rceil$.

- *Ryu's Theorem 1.4.* Ours is smaller at $k = 6$ and for every $k \ge 9$; they tie at $k = 7$, and Ryu's is smaller at $k = 8$.
- *Ryu's Theorem 1.3*, optimized over b. Ours is smaller for $k \ge 14$.
- [Fan26a]. Ours is smaller from $k = 42$ (tie at 40, 41), and larger for $15 \le k \le 39$.
- *Superseded* by Ryu's Theorem 1.2 for $k \ge 1.6 \cdot 10^7$, by his Theorem 1.1 tiers beyond that, and by his computer-assisted values: at $k = 10^5, 10^6, 10^7$ they give $c^* \le 1583, 3972, 10039$, against our $1789, 5657, 17889$.

Within $42 \le k < 1.6 \cdot 10^7$, apart from those five tabulated values, Theorem 1 is the smallest explicit bound we know of.

Acknowledgements and credit

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